

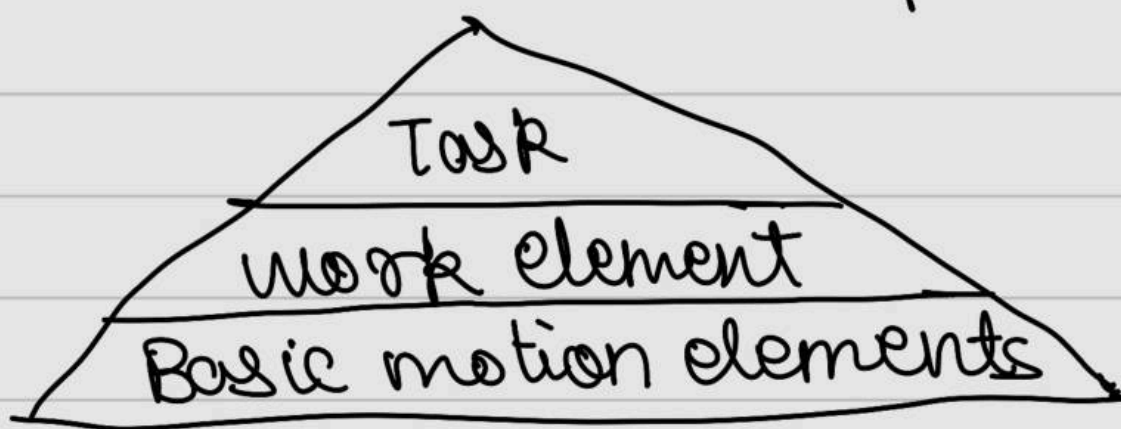
WSD Unit ①

①.0 Nature of Work

↳ Work → It is defined as an activity in which a person exerts physical and mental efforts to accomplish a given task or perform a duty.

①.1 Pyramidal struc. of Work

↳ task → amount of work that is assigned to a worker or for which a worker is responsible.

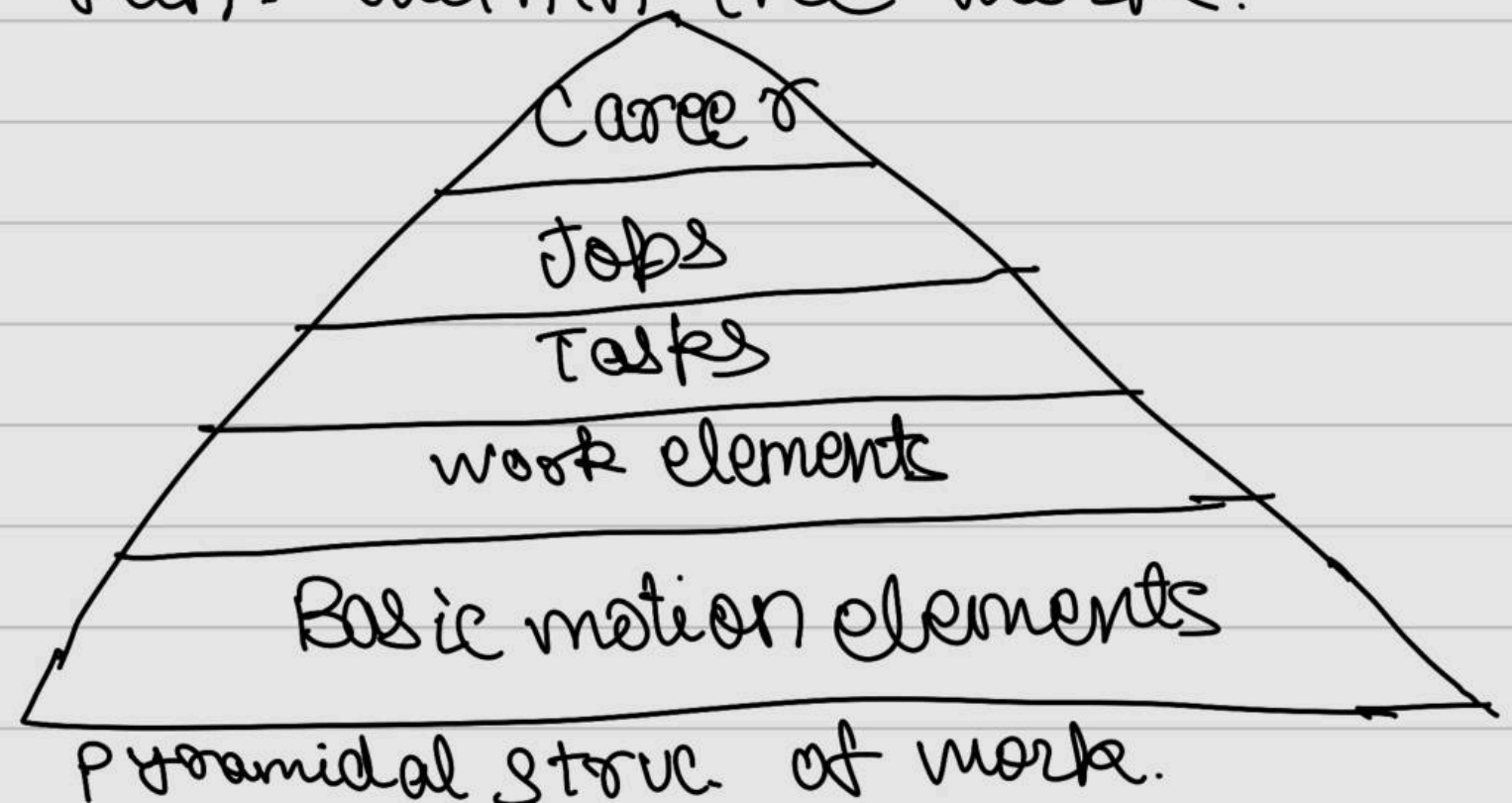


↳ Pyramidal struc. of a task. Each task consists of multiple work elements, which in turn consist of basic motion elements.

↳ Basic motion elements → are actuators of the limbs and other body parts while engaged in performing the task.

↳ Multiple basic motion elements are generally required to perform a "work element".

↳ Work element → defined as a series of work activities that are logically grouped together because they have a unified fun^c. within the work.



1.1 Importance of time

↳ Time is important in industry because of following examples

Ex: ① New product introduction

② Product cost → less work hours by workers reduce cost & saves time

③ Delivery time

④ overnight delivery

⑤ Competitive bidding.

⑥ Product scheduling.

↳ The frequently used measure of work is time.

2.0 Defining Work Systems

can be defined as -

① a physical entity and

② a field of professional practice

2.1) Physical work systems

work system $\rightarrow$ is a system consisting of humans, info and equipment that is designed to perform useful work.

$\hookrightarrow$ result of useful work is a contribution to production of a product or the delivery of a service.

$\hookrightarrow$ ex $\neq$ ① worker operating a production machine in factory

② assembly line consisting of dozen workers in diff. work stations along a moving conveyor

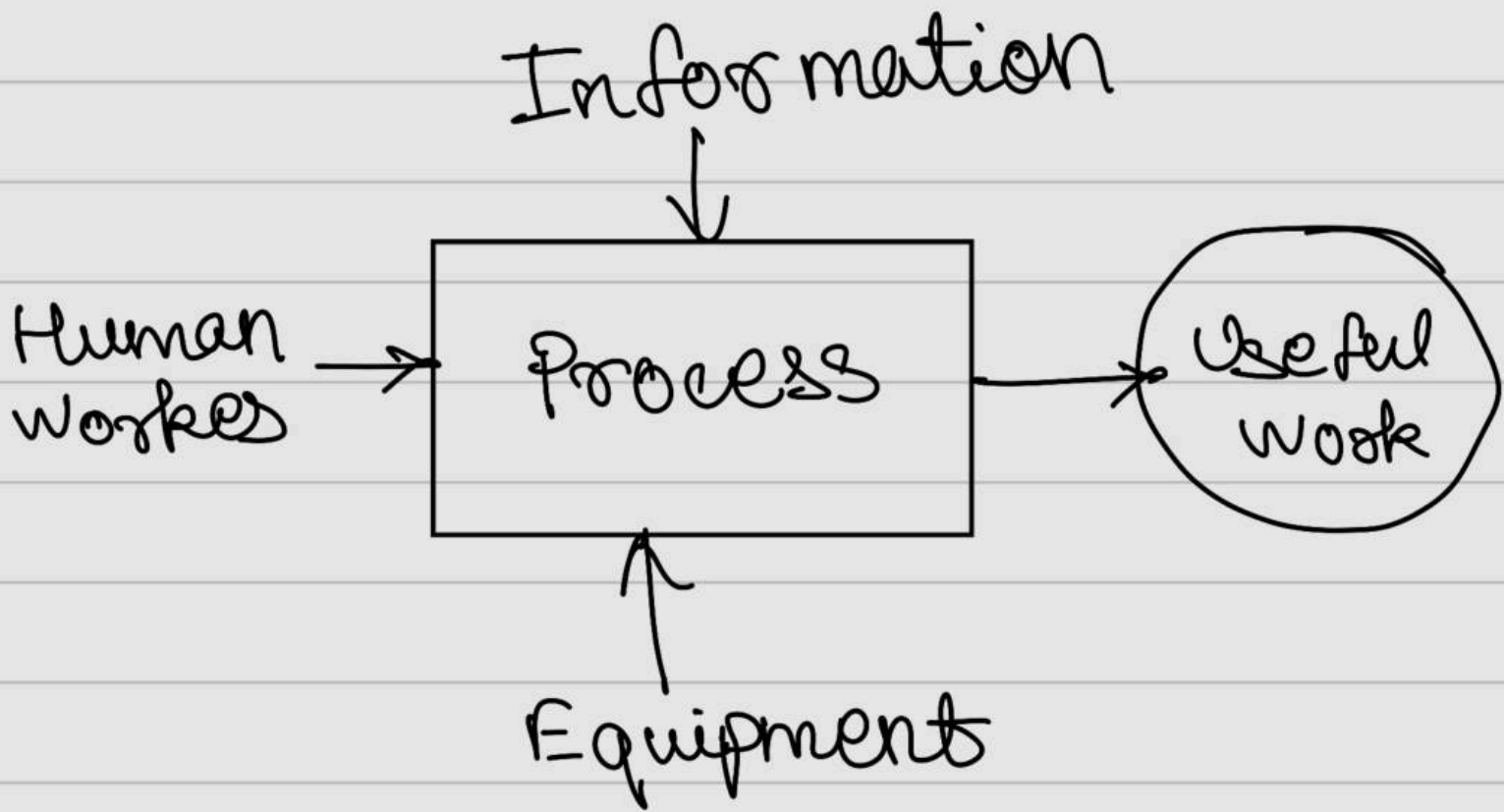


Fig → A work system consists of human workers, info., and equipment designed to accomplish useful work by means of a process.

↳ as the list indicates a work system may contain 1 or more humans, automated systems also require human attention for maintenance & reprogramming

↳ Information may include a worker's knowledge on task or a program written for computer to operate

↳ Equipment^{may} include from a single machine tool to a whole automated machine

2.1) Work system as a field of professional practise

↳ in this work system include

① Work methods

② Work measurement

③ work management

↳ term work science often used for this professional practise.

① Work methods

↳ consists of analysis and design of tasks and jobs involving human work activity

↳ Related terms operations analysis and methods engineering

↳ term motion study also used but limited to physical motions, tools and workplace layout used by a worker to perform a task

↳ other terms less restricted & include analysis and design of complex processes consisting of material and info flows through multiple operations.

② Work measurement aka time study

↳ Analysis of a task to determine the time that should be allowed for a qualified worker to perform a task, this is known as standard time (T_{std})

↳ T_{std} can be ^{also} used to compute product costs, assess worker performance & determine worker requirement.

↳ but in modern world, T_{std} has broader meaning that includes work situations in which automated machine performs the task & it is desired to determine cycle time.

↳ Time study covers any and all situations in which it is necessary to calculate the time that is req. to complete one unit of work.

③ Work management

↳ refers to various administrative and organisational tasks that are required to be complete in order to achieve high productivity of work system and effective supervision of workers.

↳ Includes functions such as-

- (i) motivating workers.
- (ii) evaluating the jobs in organisation so that each worker is paid appropriately to their work
- (iii) appraising performance of workers to reward better workers appropriately

③.0 Types of occupation (not in syllabus) broad cate.

- ↳ (a) Production workers → work involves making product.
- ↳ (b) Logistic workers → involves moving materials, products or people
- ↳ (c) Service workers → involves one of following func:
 - (i) Providing a service
 - (ii) applying existing info & knowledge
 - (iii) communicating
- ↳ (d) Knowledge workers → involves
 - (i) creating new knowledge
 - (ii) solving problems
 - (iii) managing.

4.0 Productivity

↳ defined as level of O/P of a given process relative to the level of I/P.

↳ process can refer to an individual production or service operation or it can be used in context of national economy.

↳ It is an important metric cause by improving productivity we can inc. ↑ worker compensation without inc. ↑ product & service price this leads to more products and services at lower prices for consumers, which improves std. of living for all.

④.1 Labour productivity. (most common)

$$\text{LPR} = \frac{WU}{LH}$$

LPR → labour productivity ratio

WU → work units of O/P.

LH → labour hours of I/P.

↳ fewer labour hours are better and mean higher productivity.

↳ Although labour productivity is a commonly used measure labour itself does not contribute much to improving productivity.

↳ More Imp. factors are capital & technology

↳ capital → Refers to substitution of machine for human labours

If automated replace manual machine, it inc. productivity & if no human is req. to monitor, it inc. productivity further.

↳ Technology → refers to fundamental change in the way some activity or function is accomplished.

Ex ≠ Railroad train replaced horse drawn carts.

↳ Admittedly the distinction b/w capital improvement & tech. improvements are sometimes subtle, cause new tech. almost always require capital investment.

↳ Measuring productivity is not easy as it seems, following prob. are encountered in its measurement & use. —

- Non homogeneous O/P units → O/P work

units are not necessarily homogeneous
Ex $\neq$ annual prodⁿ as O/P measure
does not account for model or if it
is a luxury car in automobile industry

- Multiple I/P factors $\rightarrow$ labour is not the only factor in determining productivity. In addⁿ to capital & tech., other factors may include materials and energy.

Ex $\neq$ In prodⁿ of Aluminium, electric power & raw material bauxite are much more imp. than labour as I/P

- Price & cost changes $\rightarrow$ Price of O/P work units and cost of I/P factors (labour, materials, power) change over time often unpredictably.

Ex $\neq$ steel industry in US during late 1990s & early 2000s

• Product mix changes $\rightarrow$ refers to the relative proportions of prod. that a company sells. If mix of exp. & Inexpensive prod. changes from year to year, an annual comparison based on LPR is less meaningful.

$\hookrightarrow$ an alternate productivity measure is labour productivity index that compares the O/P-I/P ratio from one year to next.

$$LPI = \frac{LPR_t}{LPR_b}$$

LPR_t = Labour Productivity ratio during some time period of interest
 LPR_b = LPR during some defined base period.

Ex # During a base year in a small steel mill, 326,000 tons of steel were produced using 203,000 labour hours. In next year, the O/P was 341,000 tons using 246,000 labour hours. Determine (a) labour productivity ratio for the base year (b) the labour productivity ratio for second year (c) labour productivity index for second year.

Solⁿ → (a) In Base year,
$$LPR = \frac{326,000}{203,000} = 1.6059 \text{ tons/hrs}$$

(b) In next year,
$$LPR = \frac{341,000}{246,000} = 1.3861 \text{ tons/hrs}$$

(c) LPI for 2nd year using base year
$$LPI = \frac{1.3861}{1.6059} = 0.8631$$

→ Productivity went down in second year

④.2 Productive work content

↳ Work is usually not performed in most efficient way possible, given task performed by worker can be considered to consist of

- (i) the basic productive work content
- (ii) excess non productive activities

(i) is the theoretical min^m. amount of work req. to accomplish the task, where the amount of work is expressed in terms of time. Thus the time required to perform the basic productive work content is the theoretical minimum. It cannot be further reduced.

(ii) are the extra physical and mental actions performed by the worker that do not add

any value to the task, nor do they facilitate the productive work content that does add value. Add to basic productive work content. time to make total time needed to perform the task

↳ can be classified into three -
① caused by poor design of Product
② caused by inefficient methods, poor work layout & interruptions
③ caused by human factor.

↳ nonproductive activities do not necessarily occur during every cycle of task. Ex: accidents.

Basic productive work content	Excess activities due to poor design of product	Excess activities due to poor work layout, insufficient methods	Excess due to human factor
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Productive
time

Total excess time

Total task time

